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Pandeeswaran Munisamy
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Kalaimani Subramani

THE PREPARATORY BOOK OF ENVIRONMENTAL STUDIES

NUT CELLS OF ECOLOGY(EVS)



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THE PREPARATORY BOOK OF ENVIRONMENTAL STUDIES

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FOREWORD



Prof. Dr. M. Srinivasan, Ph.D.

Science Advisor,

Bharath Rhino Biotech

The quality of human life in the earth depends on the quality of air, water, food and shelter that provided by the Mother Nature. Natures also made a law that regulate all life that exist in our earth with inter dependent service network system that is scientifically called ecosystem. Anything that violates this law will be punished by the Mother Nature. Some past history reveals that mass extinction happened due to severe punishment. Hence, every creature in the earth must understand this ecosystem and associated law of nature. This book is prepared by authors with neat illustrations that obviously enable the readers, even if, just spend a minute for a glance. All literature having scientific facts with selective title reviewed are more appropriate. I hope the skillful narration of the book topics and systematic approach dealt with ENVIRONMENTAL SCIENCE that published by GTN Arts College, Dindigul, Tamilnadu, India is a great contribution to college level academic institutions and other societies involve in the environmental studies and Eco-care.

Thanjavur

March-2022

A handwritten signature in blue ink, appearing to read 'M. Srinivasan', written over a horizontal line.

Prof. Dr. M. Srinivasan, Ph.D.

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Words find no bounds to express my sincere thanks to all who made this Allied practical manual "THE PREPARATORY BOOK OF ENVIRONMENTAL STUDIES" to come true.

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I thank Dr.A.PANDIARAJAN, Assistant Professor and Head, PG and Research Department of Chemistry, GTN College of Arts and Science, Dindigul.

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I wish to express my profound thanks to All the Staff Members, Department of chemistry, GTN College of Arts and Science, Dindigul.

And last but not least I wish to extend my sincere thanks to All My Friends and Relatives for their co- operation and support in this Book to make success.

Lt.Dr.M.PANDEESWARAN,

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UNIT -1

ENVIRONMENT EDUCATION:

INTRODUCTION:-

Environmental education enables learners to develop a structure of knowledge about the world and seek knowledge that they can use and develop throughout their lives. Environmental education empowers learners by enabling them to participate in a sustainable future. Thus the foundation for a lifelong learning is laid by environmental education.

MEANING OF ENVIRONMENTAL EDUCATION:-

Environment is derived from the French word “Environner”, which means encircle or surrounding. Environment is a complex of many variables, which surrounds man as well as the living organisms. Environmental education describe the interrelationships among organisms, the environment and all the factors, which influence life on earth, including atmospheric conditions, food chains, the water cycle, etc. It is a basic science about our earth and its daily activities, and therefore, this science is important for everyone.

SCOPE OF ENVIRONMENTAL EDUCATION:-

Environmental education discipline has multiple and multilevel scopes. This study is important and necessary not only for children but also for everyone. The scopes are summarized as follows:

1. The study creates awareness among the people to know about various renewable and nonrenewable resources of the region. The endowment or potential, patterns of utilization and the balance of various resources available for future use in the state of a country are analyzed in the study
2. It provides the knowledge about ecological systems and cause and effect relationships.
3. It provides necessary information about biodiversity richness and the potential dangers to the species of plants, animals and microorganisms in the environment.
4. The study enables one to understand the causes and consequences due to natural and induced disasters (flood, earthquake, earthquake, landslide, cyclones etc.) and pollutions and measures to minimize the effects.
5. It enables one to evaluate alternative responses to environmental issues before deciding an alternative course of action.
6. The study enables environmentally literate citizens (by knowing the environmental acts, rights, rules, legislation, etc.) to make appropriate judgments and decisions for the protection and improvement of the earth.
7. The study exposes the problems of over population, health, hygiene, etc. and the role of arts, science and technology in eliminating/ minimizing the evils from the society.
8. The study tries to identify and develop appropriate and indigenous eco-friendly skills and technologies to various environmental issues.
9. It teaches the citizens the need for sustainable utilization of resources as these resources are inherited from our ancestors to the younger generation without deteriorating their quality.

10. The study enables theoretical knowledge into practice and the multiple uses of environment.

IMPORTANCE OF ENVIRONMENTAL EDUCATION:-

Environmental study is based upon a comprehensive view of various environmental systems. It aims to make the citizens competent to do scientific work and to find out practical solutions to current environmental problems. The citizens acquire the ability to analyze the environmental parameters like the aquatic, terrestrial and atmospheric systems and their interactions with the biosphere and atmosphere.

1. World population is increasing at an alarming rate especially in developing countries.
2. The natural resources endowment in the earth is limited.
3. The methods and techniques of exploiting natural resources are advanced.
4. The resources are over-exploited and there is no foresight of leaving the resources to the future generations.
5. The unplanned exploitation of natural resources lead to pollution of all types and at all levels.
6. The pollution and degraded environment seriously affect the health of all living things on earth, including man.

7. The people should take a combined responsibility for the deteriorating environment and begin to take appropriate actions to save the earth.
8. Education and training are needed to save the biodiversity and species extinction.
9. The urban area, coupled with industries, is major sources of pollution.
10. The number and area extinct under protected area should be increased so that the wild life is protected at least in these sites.
11. The study enables the people to understand the complexities of the environment and need for the people to adapt appropriate activities and pursue sustainable development, which are harmonious with the environment.
12. The study motivates students to get involved in community action, and to participate in various environment and management projects.
13. It is a high time to reorient educational systems and curricula towards these needs.
14. Environmental education takes a multidisciplinary approach to the study of human interactions with the natural environment.
15. Environmental study is a key instrument for bringing about the changes in the knowledge, values, behaviors and lifestyles required to achieve sustainability and stability within and among countries.
16. Environmental education deals with every issue that affects an organism. It is essentially a multidisciplinary approach that brings about an appreciation of our natural world and human impacts on it integrity.
17. Practical answers to making human civilization sustainable on the earth's finite resources.

GUIDING PRINCIPLES OF ENVIRONMENTAL EDUCATION:

1. Consider the environment in its totality, natural and built technological and social structures (economic, political, technological, cultural, historical, moral and aesthetic).
2. Environmental education to be a continuous life saving process (beginning at the pre-school level continuing through all formal and non-formal stages).
3. Environmental education to be interdisciplinary in its approach.
4. Examine major environmental issues from local, national and international point of view.
5. Environmental education to focus on current and potential environmental situations.
6. Promote the values and necessity of local, national and international cooperation in the prevention and solution to environmental problems.
7. Explicitly consider environmental aspects of plan for development and growth.
8. Enhance the position of learners in making decision concerning their environment and accept responsibility.
9. Enable learners to discover symptoms and real and potential causes of environmental problems.
10. Enhance the learners' ability to develop critical thinking and problem solving skills.

11. Utilize different learning environment and approaches to learning/teaching about and form of the environment with emphasis on first hand information.

OBJECTIVES OF ENVIRONMENTAL EDUCATION AT SCHOOL

i. Objectives of Environmental Education at Primary Level

To know and understand true aspects of the environment in general.

- (a) To know and understand the interaction between mammals, between human and their environment and interaction between the various elements and components of the environment.
- (b) Build understanding, awareness and sensitivity towards causes and efforts of the class that continuously take place in society the world around us.
- (c) To build and develop skills in thinking, reasoning, enquiring, evaluating and making decisions concerning human and the world around them.
- (d) Inculcate the attitude in using the knowledge and skills towards solving problem and issues related to individuals, society and the environment.
- (e) To build the values and attitudes towards the need and necessity to live together in harmony in the context of the heterogeneous society.

Focus

Human, animal and plants undergo number of life -processes.

1. Human, animals and plants are continuously adapting themselves to the environment.
2. Human alters and modifies the environment with great caution and care in order to fulfill numerous living needs.
3. Identification between human and nature and between environmental elements giving rise to various phenomena which affect them.
4. Society would take active steps to conserve the environment and the balance of nature through careful plans and processing. Objectives of Environmental Education at Secondary Level
5. Environmental education to be taught as integrated science in which environmental education concepts are included.

OBJECTIVES

1. To emphasize the relevance of science to daily life.
 2. To develop a scientific attitude in student.
 3. To create an environment conducive to greater reliance on the use of principles and practice of science.
 4. To acquaint the student's with various natural phenomena.
 5. To develop an outlook which emphasizes the method employed in different disciplines of science.
- ii. Aspects of Environmental Education Emphasized at Higher Secondary Level
- (a) Population - growth, rises and problems of unplanned population.

- (b) Law - Land use, land reclamation and land and soil conservation.
- (c) Resources - resource uses, conservation, recycling.
- (d) Food and Nutrition - Food production, food adulteration and preservation, balance diet etc.
- (e) Conservation - Causes of wildlife, plant, soil, water and conservation of other non-renewable natural beauty.
- (f) Pollution - Pollution of water, air and soil, noise pollution, pollution by insecticide and other chemicals and waste disposals.
- (g) Health and Hygiene - Individual, family, country and social health and hygiene, health hazards etc.
- (h) Humans and Nature - Other compounds of atmosphere, environmental quality and future on earth.

CONSTRAINTS TO IMPLEMENTING ENVIRONMENTAL EDUCATION:

1. . Rigid Specialization.
2. Complexity of inter-disciplinary value of Environmental education.
3. High pupil - teacher ratio for organizing pupil participation programs.
4. Paucity of qualified trained environmental educator.
5. Lack of proper resources in terms of equipment, supplementary materials and reference materials.
6. Tendency to resist changes

ENVIRONMENT SCIENCE:-

- Environment science is the study of the environment, its biotic (i.e., biological) and a biotic (i.e. non-biological) components and their interrelationship.
- Environmental studies are the process of educating the people for preserving quality environment.

TYPES OF ENVIRONMENT:

Environment can be divided into two categories are Natural environment and Man– made environment.

1.Natural environment

Natural environment is characterized by natural components. All biotic (living) and biotic components (non-living) are created through a natural process. Creation of these biotic and abiotic components does not require any human support. Example: soil, water, air, trees, radiations, noise, etc.

2.Man – made environment

Man is the most powerful environmental agent. He modifies the environment using modern technologies, according to his needs largely. Thus, man creates the man- made environment. Example: house, road, schools, railway lines, parks, etc.

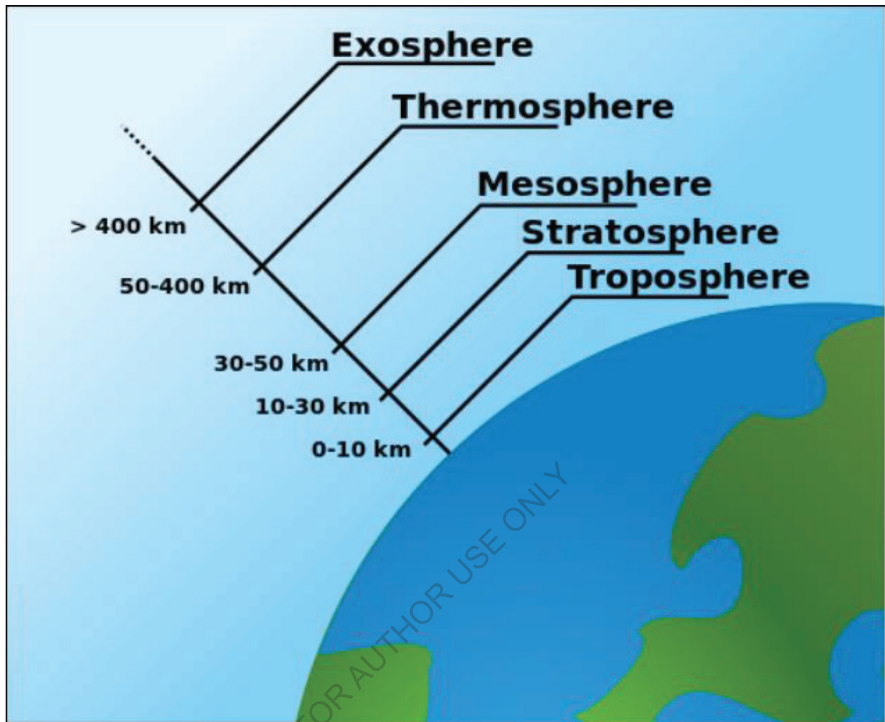
3.Earth

We live on the planet. All the Factors necessary for life are present only on the Earth. No other planet in the solar system can sustain life. The average diameter of the Earth is about 12,800 kilometers. The earth takes 365 1/4 days to complete one revolution round the sun. It also rotates on its own axis once in 24 hours.

SURVIVAL OF LIFE ON EARTH:

Our Earth is the only planet of the solar system where life originates and survives. It is also at the right distance from the sun to retain water. This water is able to absorb the excess carbon dioxide and retains it on the Earth in the form of carbonate rocks. Thus, the temperature of the earth's surface remains to a comfortable level due to the removal of carbon dioxide. It makes the Earth a suitable planet for the life to originate and survive. The following factors also support the case.

1. The Earth has essential elements like carbon, hydrogen, nitrogen and oxygen, which act as building blocks for the evaluation of the life.
2. The Earth has sufficiently large amount of water in the form of lakes, Rivers and oceans for growth and survival of life.
3. The Earth has the suitable temperature for carrying out the sustaining chemical reactions. This temperature is neither too high nor too low.
4. The Earth also has a protective blanket of ozone layer in the atmosphere to save life from harmful effects of ultra violet-radiations coming from the sun.



DIFFERENT PARTS OF THE EARTH:-

Earth can be divided into four parts are Lithosphere, Atmosphere, Hydrosphere and Biosphere

I. ATMOSPHERE:

The cover of air, that envelope the earth is known as the atmosphere. The atmosphere extends up-to 500 kms from the earth surface. The atmosphere is essential for all living organisms. It comprises 78% of nitrogen, 21% of oxygen and 1% of other gases.

STRUCTURE OF ATMOSPHERE:

Structure of atmosphere can be classified into the following three types.

- a. Troposphere (10 – 8 kms): it contains 75% of the atmosphere air mass. It also contains moisture.
- b. Stratosphere (18–50 kms): it is rich in ozone gas and free from moisture and clouds. prevents the UV radiation from the sun
- c. Mesosphere (50 – 85 kms): it contains less ozone, but more nitrogen oxide.

FUNCTIONS OF ATMOSPHERE:

1. It maintains the heat balance on the earth by absorbing the IR radiations.
2. The gaseous constituents play an important role in sustaining life on earth.

GASEOUS CONSTITUENT	FUNCTIONS
Oxygen	Supports life of living organisms.
Carbon-dioxide	Essential for photosynthetic activity of Plants.
Nitrogen	Essential nutrient for plant Growth.

II.LITHOSPHERE

The soil and rock components of the earth called Lithosphere. **Functions of lithosphere:**

1. It is a home for human beings and wildlife.
2. It is a storehouse of minerals and organic matters.

III.HYDROSPHERE

The aqueous envelope of the earth (i.e., 75% of the earth surface) is called hydrosphere. Oceans, lakes, streams, rivers and water vapors constitute hydrosphere. About 97% of earth's water is in oceans, which is too salty and not fit for drinking. Only 3% is available as fresh water.

Functions of hydrosphere

1. It is used for drinking purpose and supports the aquatic life.
2. It is also used for irrigation, power production, industries and transport.

III.BIOSPHERE:

The biological environment, where the living organisms live and interact with physical environment (soil, water and air) is called biosphere.

Functions of biosphere

1. Plants through photosynthesis produce oxygen in the atmosphere.
2. Animals inhale oxygen during respiration and give out carbon dioxide, which is again utilized by plants during photosynthesis.

GLOBAL ISSUES:

OZONEDEPLTION:

Ozone Layer:

- ☐ Ozone layer acts as a filter around the earth and shields harmful rays such as UV reaching the earth thus protecting the earth. The ozone shield protecting the earth from dangerous ultra violet region.
- ☐ 90% of ozone lies in the stratosphere, which is located in between 15-50 km above the earth surface..

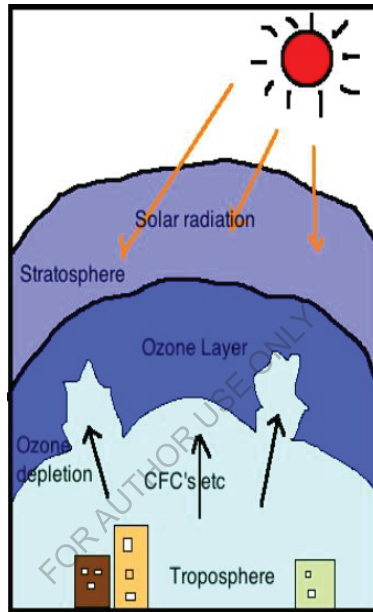
Causes of Ozone Depletion:

- ☐ The main cause of ozone depletion is the widespread use of Chlorofluoro carbons (CFCS)
- ☐ CFC are used as, coolant in air conditioners, refrigerants, deodorants, hairsprays, shaving creams accumulate in the stratosphere.
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Effects of ozone depletion;

- ☐ The harmful U.V radiation towards earth increases UV induces change in DNA, which is responsible for skin cancer.
- ☐ Crops (aquatic and terrestrial ecosystem) would be adversely affected (crop yield and quality are reduced).

- Affects quality of air, splitting of Ozone that will affect human health, plants and outdoor materials.
- It causes health hazards in human, animals and wetland plants.



GLOBAL WARMING:

Definition

Abnormal increase of the temperature in the environment due to the continuous accumulation of gaseous pollutants is known as global warming.

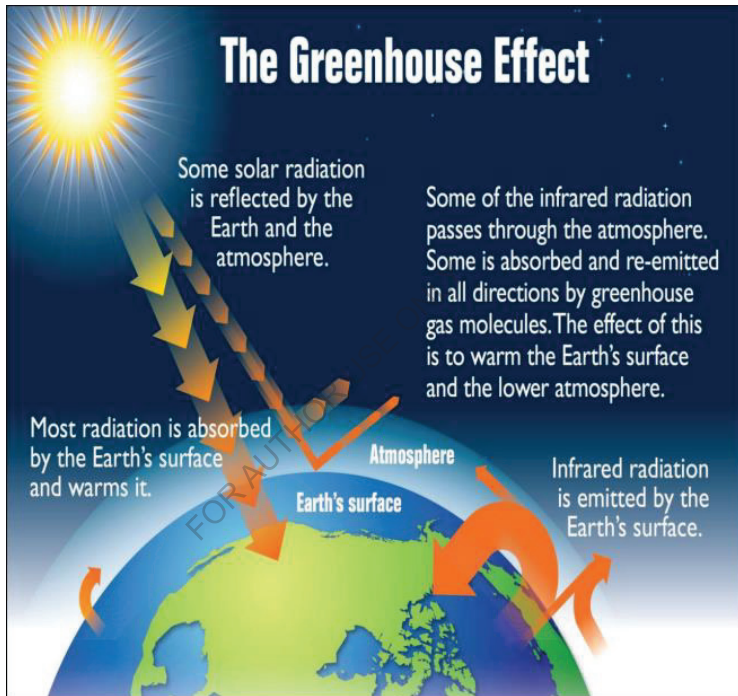
Greenhouse effect:

- It is also defined as the progressive warming up the earth's surface due to blanking effect of manmade CO₂ in the atmosphere.
- Greenhouse means that it allows Short wave incoming solar radiation but it does not allow the long wave outgoing infra red radiation to escape.
- We can feel the effect of green house by taking example of car. All the glass doors of the car is closed.
- The car is parked in a hot summer day. After a while, if we go inside a car, it will be very hot and it is hotter than the outside. I.e. sunlight is passing through the glasses and is warming up the seats.
- The heat radiated by the seats and inner parts are not allowed to escape through the glass. Hence the car inside is better than the outside atmosphere. It is because of the green house effect.
- In a similar way, the earth's atmosphere acts like a greenhouse. Carbon dioxide in the atmosphere transmits incoming shortwave length solar radiation (sunlight) but does not transmit outgoing long wavelength heat radiation from the warmed surface of the earth.
- Instead, these long wavelength heat radiations are reflected on the earth repeatedly and this is known as Green house effect.

Green gases:

□ The adverse effects of green house are caused by the heat trapping gases like carbon dioxide (CO₂) methane (CH₄). Nitrous oxide (NO₂) and Chlorofluoro carbons (CFCS). These are all called green house gases. Among these, CO₂ is the most common and important green house gas.

□ The gases like (CO_2), (NO_2), (CH_4), O_3 and Chlorofluoro carbons (CFC) present in the atmosphere are absorbing long wave radiation and radiate energy back to earth. When these increases in the atmosphere because of air pollution by human activities, more heat is radiated back to earth and consequently temperature of the earth increases. This phenomenon is known as greenhouse effect (or) global warming.



Effects of global warming:

- This increased temperature may lead to melting of polar icecaps, releasing huge quantities of water into oceans.
- This may cause heavy tidal waves and even submerge many coastal cities of the world, destroying millions of lives. A mere like India having huge coastline, its coastal population is especially vulnerable to the dangers of global warming.

The most effective way of preventing global warming is to plant more trees.

LOSS OF BIODIVERSITY:

Definition:

The totality of genes, species and ecosystems in a defined area.

Loss of biodiversity definition

It refers to either the ongoing extinction of species at a global level or the local reduction or loss of species in a given habitat.

The scale of the problem:

In the last 40 years, we have lost 52% of planetary biodiversity & lost 58% of vertebrates on land, sea and air - the vertebrate figure could rise to 66% by 2020.

Natural causes of biodiversity loss

Natural forces can act to destroy habitat, species and individual organisms. Obvious examples include volcanic eruptions, floods and fire. Previous mass extinction events have been associated with asteroid impact.

Man-made causes of biodiversity loss

1.Climate change:

Induced through man-made activities although to provide a balanced view, species can gain environmental advantage or lose it when the climate changes.

3. Pollution;

On Land, Air and Water. Water systems suffer aquatic nutrient load from fertilizers and agricultural by-products. Oceans are seeing rising acidity levels caused by man-made pollutant activity.

Loss of biodiversity:

1. The farmers prefer hybrid seeds; as a result, many plant species become extinct.
2. For the production of drugs, the pharmaceutical companies collect wild plants, so several medicinal plants now become extinct.
3. Tropical forest is the main sources of world's medicine. Every year these forests are disappearing due to agriculture, mining and logging.

ACID RAIN

It is a rain that is unusually acidic, meaning that it possesses elevated levels of hydrogen ions. It can have harmful effects on plants, aquatic animals and infrastructure. Acid rain is caused by emissions of sulfur dioxide and nitrogen oxide, which react with the water molecules in the atmosphere to produce acids. Some Governments have made efforts since the 1970s to reduce the release of sulfur dioxide and nitrogen oxide into the atmosphere with positive results.

Nitrogen oxides can also be produced naturally by lightning strikes, and sulfur dioxide is produced by volcanic eruptions. The chemicals in acid rain can cause paint to peel, corrosion of steel structures such as bridges, and weathering of stone buildings and statues.

1. Natural phenomena

The principal natural phenomena that contribute acid-producing gases to the atmosphere are emissions from volcanoes. Volcanoes create extremely high amounts of acid rain and fog, with acidity as high as a pH of 2, clearing an area of any vegetation and frequently causing irritation to the eyes and lungs of inhabitants in nearby settlements.

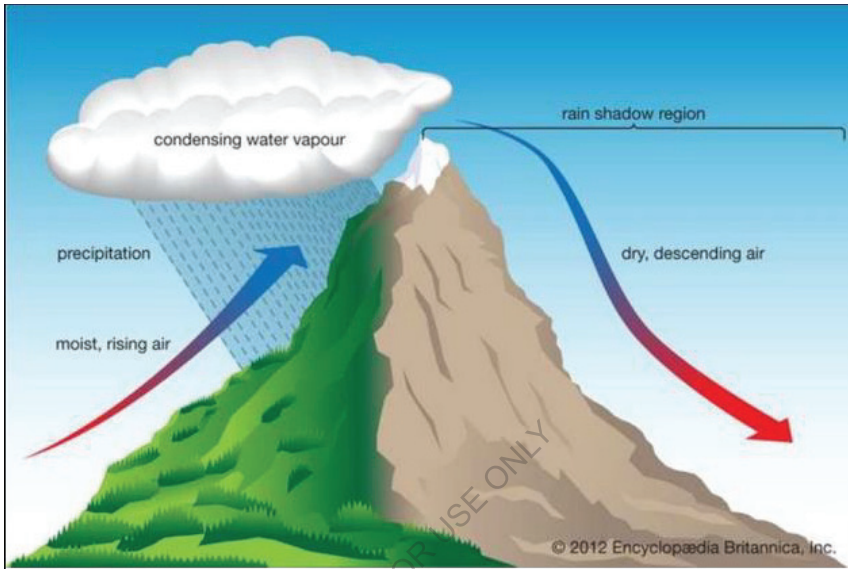
2. Human activity

The principal cause of acid rain is sulfur and nitrogen compounds from human sources, such as electricity generation, factories, and motor vehicles. Electrical power generation using coal is among the greatest contributors to gaseous pollution that are responsible for acidic rain. The gases can be carried hundreds of kilometers in the atmosphere before they are converted to acids and deposited. In the past, factories had short funnels to let out smoke but this caused many problems locally; thus, factories now have taller smoke funnels. However, dispersal from these taller stacks causes pollutants to be carried farther, causing widespread ecological damage.

3. Cloud Burst:

Cloudburst, a sudden, very heavy rainfall, usually local in nature and of brief duration. Most so-called cloudbursts occur in connection with thunderstorms. In these storms there are violent uprushes of air, which at times prevent the condensing raindrops from falling to the ground. A large amount of water may thus accumulate at high levels, and if the upward currents are weakened the whole of this waterfalls at one time.

Cloudbursts are especially common in mountainous areas. This is probably because the warm air currents of a thunderstorm tend to follow the upward slope of a mountain. The effects of heavy rain are especially striking on mountain slopes because the falling water is concentrated in valleys and gullies. Mountain cloudbursts cause sudden and destructive floods. The intensity of rainfall in the most severe cloudbursts can only be conjectured. There have been cases, however, in which the excavations made in the ground by the falling water of a cloudburst appear to indicate an even greater intensity of rainfall.



DESERTIFICATION;

Desertification, also called Desensitization, the process by which natural or human causes reduce the biological productivity of dry lands (arid and semiarid lands). Declines in productivity may be the result of climate change, deforestation, overgrazing, poverty, political instability, unsustainable irrigation practices, or combinations of these factors. The concept does not refer to the physical expansion of existing deserts but rather to the various processes that threaten all dry land ecosystems, including deserts as well as grasslands and scrublands.

Causes:

Desertification is caused by variations in climate and by unsustainable land-management practices in dry land environments. By their very nature, arid and semiarid ecosystems are characterized by sparse or variable rainfall. Thus, climatic changes such as those that result in extended droughts can rapidly reduce the biological productivity of those ecosystems. Such changes may be temporary, lasting only a season or they may persist over many years and decades. On the other hand, plants and animals are quick to take advantage of wetter periods, and productivity can rapidly increase during these times.

E-WASTE (ELECTRONIC AND ELECTRICAL WASTES):

Electronic equipment like computers, printers, mobile phones, Xerox, machines, calculators, etc., after using these instruments, they are thrown as waste.

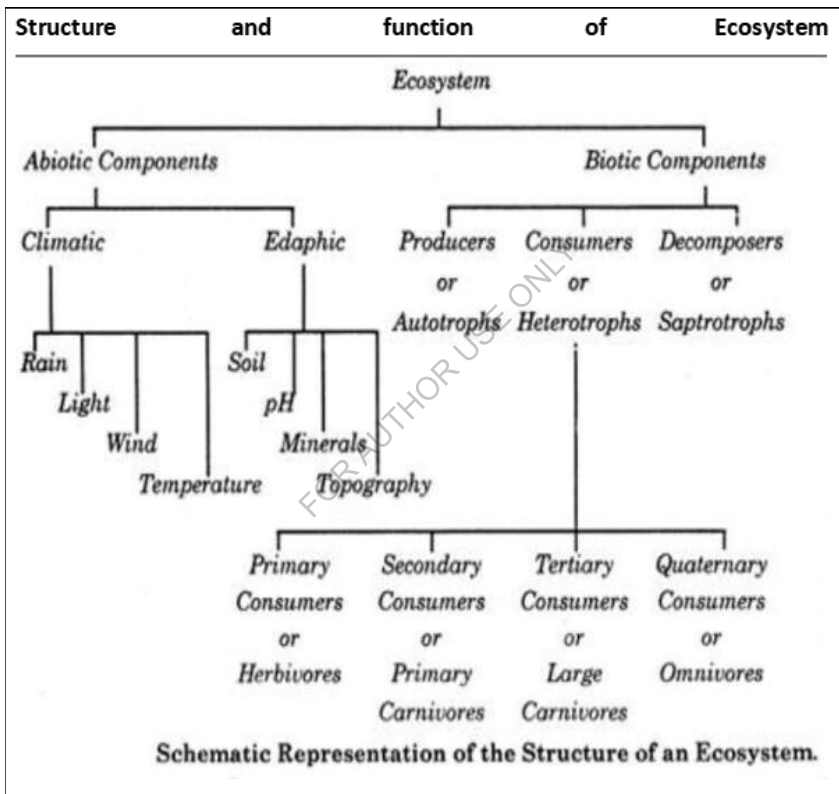
- People discard computers every two to four years on average.
- Cell phones have a life-cycle of less than two years in industrialized countries.
- Each computer screen contains about 20% lead by weight.
- A mobile phone is 19 % copper and 8%iron.
- Informal name for electronic products nearing end of their “useful life”.
- Large household appliances - Refrigerators Air conditioners, computers & Stereo systems, Mobile phones.

- Its volume increases by 3-5% per annul.
- Major pollutants are Heavy metals – Hg, Pb, Cd, Cr (VI) and Flame retardants – Polybrominated biphenyls (PBB) and Polybrominated biphenyl ethers (PBDEs).

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UNIT - II ECOSYSTEM

Ecosystem is the basic functional unit of ecology. The term ecosystem is coined from a Greek word meaning study of home. Ecosystem is a self-sustaining structural and functional unit of biosphere. It consists of living beings and the physical environment, both interacting and exchanging materials between them.



An ecosystem has been classified into biotic (living) components and abiotic (non-living) components.

1. Biotic components:

They include the living components of an ecosystem. They can be put in the following categories. *i.Producers, ii.Consumers, iii.De composers.*

(i) Producers (Autotrophs)

Producers synthesize their food themselves through photosynthesis.

Example: all green plants, grasses and trees.

(ii) Consumers (heterotrophs)

Consumers are mostly animals. This cannot prepare their own food and depend directly or indirectly on the producers for food. *Examples: rabbit, cow, lions, tiger, etc.*

Classification of consumers

Consumers are further classified as

a) Primary consumers (Herbivores) (plant eaters)

Primary consumers are also called herbivores; they directly depend on the plants for their food. Therefore, they are called plant eaters. *(e.g.) rabbit, goat, cow, horse, etc.*

b) Secondary consumers (Primary Carnivores) (meat eaters)

Secondary consumers are primary carnivores; they feed on primary consumers. They directly depend on the herbivores for their food. *(e.g.)Frog, cat, snakes, etc.*

c) Tertiary consumers (Secondary Carnivores) (meat eaters)

Tertiary consumers are Secondary carnivores; they feed on Secondary consumers. They directly depend on the primary carnivores for their food. *(e.g.) lion, tiger etc.*

d) Quaternary consumers (Omnivores)

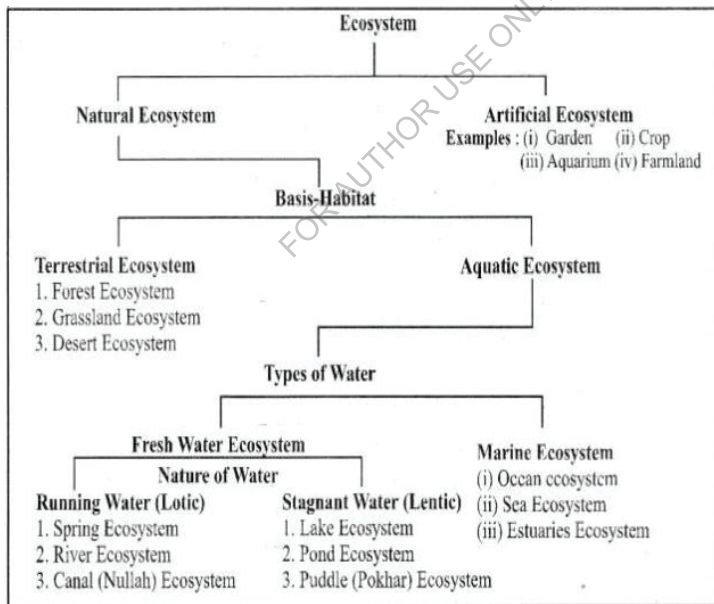
These organisms obtain their food by feeding on both plants and animals
(e.g.) *humans, cockroaches, fox etc.*

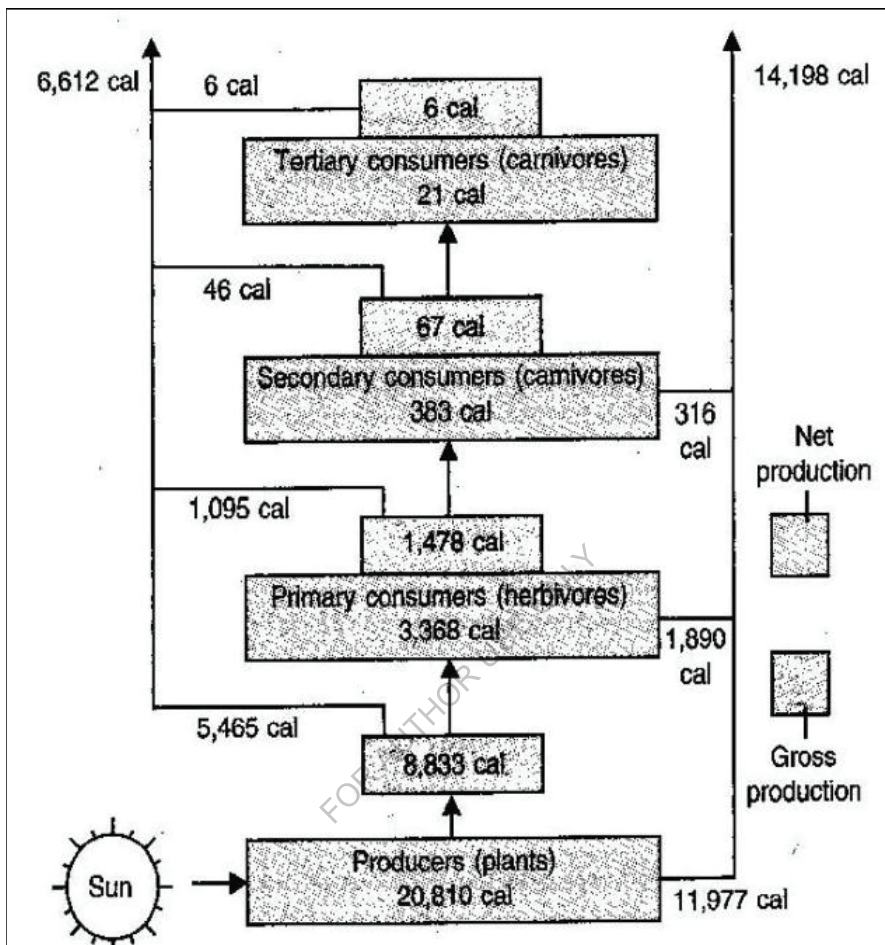
(iii) Decomposers

Composers attack the dead bodies of plants and animals and break them into simple components. During the decomposition, inorganic nutrients are released. The producers for the synthesis of their own food then utilize these inorganic nutrients together with other organic substances. (e.g.) *Microorganisms like fungi and bacteria.*

(b) Abiotic components

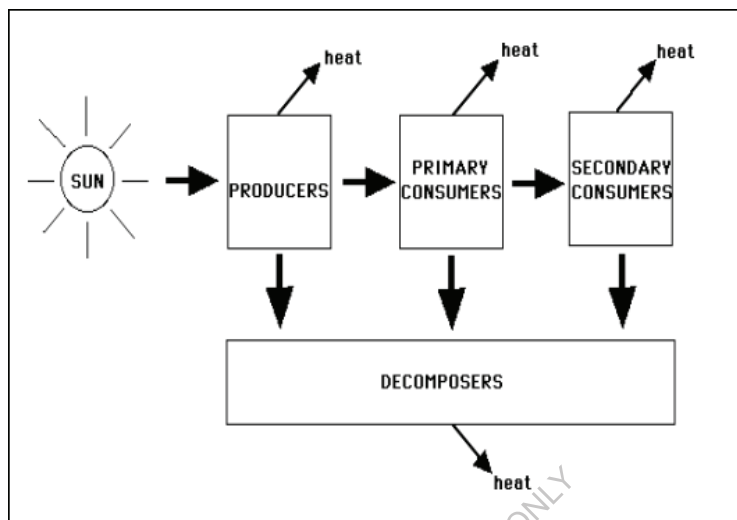
It is the non-living component of an ecosystem and includes physical components and chemical components.





Energy flow in the ecosystem

The maintain life activities-energy is required. The plants make use of water, salts and CO_2 from the environment to prepare food namely starch. This synthesis is aided by the energy derived from the sunlight trapped by chlorophyll. Thus, energy from the sun enters the world through photosynthetic organisms and passes on from one organism to another in the form of food.



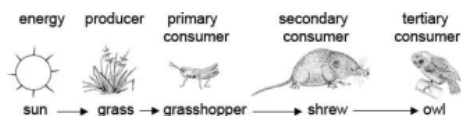
The energy flow in the ecosystem is unidirectional [one-way traffic] and non-cyclic. Laws of thermodynamics, which state that, govern this one-way flow of energy and are neither created nor destroyed but may be transferred from one form to another.

During energy transfer, there is a degradation of energy from a concentrated form [mechanical chemical (or) electrical etc] to a dispersed form [heat].

Thus, there is a continuous loss of energy within each trophic level from producer to consumer within an ecosystem. This loss of energy is dissipated as heat during metabolism of the organisms present in the food chain. It also indicates that shorter the food chain, greater would be the available food energy, and with an increase in the length of the food chain, there is a corresponding increase in loss of energy.

Food chain

The food energy is transferred from the producers, through a series of organisms-herbivores, carnivores and de composers. *The sequence of eating and being eaten in an Ecosystem is known as Food chain.*



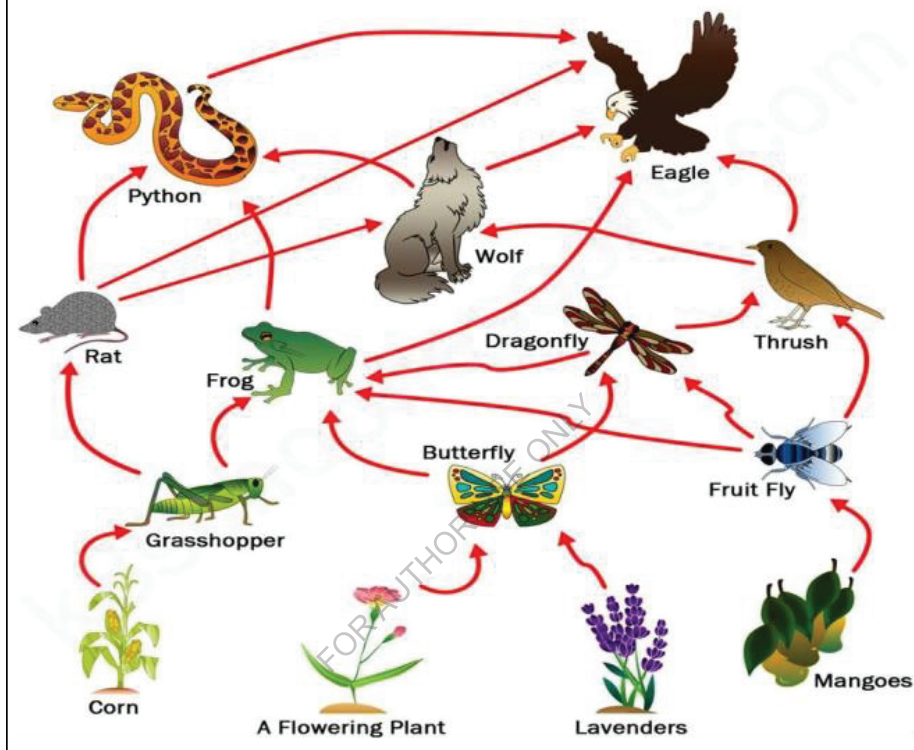
In simple words, food chain indicates who eats who in an ecosystem.

- (i) A food chain always begins with a producer followed by herbivores and then carnivores.
- (ii) At each step, lot of energy from food is lost as heat. Therefore, the number of steps in any food chain is limited to four or five.
- (iii) In a food chain, each step represents a trophic level.

Food web

- The interlocking pattern of various food chains in an ecosystem is known as food web.
- The linear arrangement of food chains hardly occurs and they are interlocking connected under natural condition and hence they are found alternatives in nature. In grazing food chains of grassland, in the absence of rabbit, grass may be eaten by mouse.
- The mouse may be eaten by directly hawk (or) by snake first and then the snake is eaten by hawk.

A Food Web



Example:

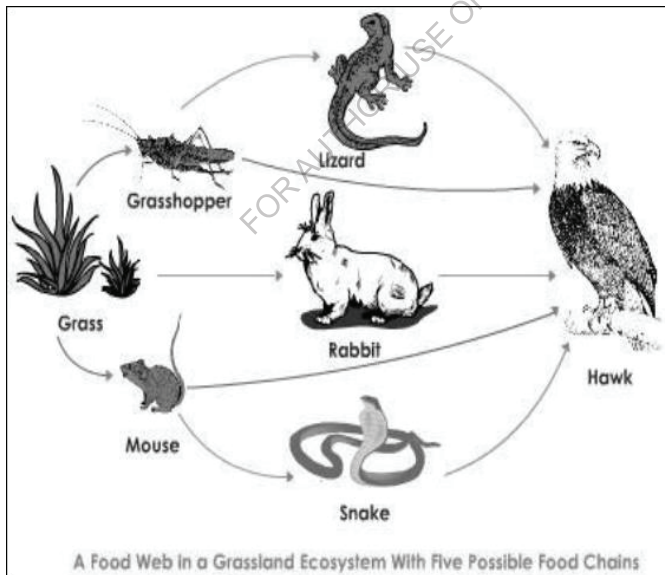
There are five linear food chains, which are interconnected to form a food web.

1. Grass ---> Grasshopper---> Hawk
2. Grass ---> Grasshopper---> Lizard ---> Hawk

3. Grass ---> Rabbit ---> Hawk
4. Grass ---> Mouse ---> Hawk
5. Grass ---> Mouse --->snake ---> Hawk

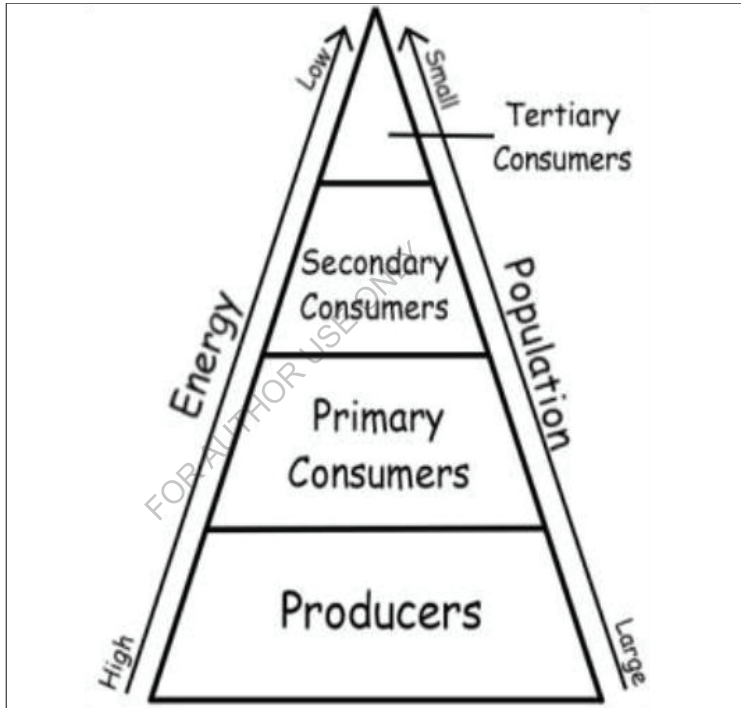
Grasshopper, rabbit and mouse eat Grass. Grasshopper is eaten by lizard, which is eaten by hawk. Hawk eats rabbit. Mouse is eaten by snake, which is eaten by hawk. In addition, hawk also directly eats grasshopper and mouse.

The food web provides more than one alternatives of food of most of the organisms in an ecosystem and therefore, increases their chances of survival.



Ecological Pyramids

Ecological pyramid is the graphical representation of the number, biomass and energy of the successive trophic levels of an ecosystem. In the ecological pyramid, the producer forms the base and the final consumer occupies the apex.



Types of ecological pyramids

Ecological pyramids are classified into *Pyramid of number*, *Pyramid of biomass* and *Pyramid of energy*.

1. Pyramid of Number

It shows the number of individual organisms at each trophic level. For example, in a Grassland ecosystem, the producers are mainly grasses. Which are small and large in numbers so the producers occupy lower trophic level (1st Trophic level).

The primary consumers (herbivores) like rabbit, cow, deer, grasshopper, etc. are lesser in number than grasses. Which occupy the 2nd trophic level the secondary consumers snakes and lizards are still lesser in number than herbivores. Which occupy the 3rd trophic level finally, the tertiary consumers like hawks, lions are least in number. Which occupy the next trophic level. Therefore, the pyramid becomes upright.

2. Pyramid of Biomass

It represents the total amount of Biomass (mass or weight of biological organism) present in each trophic level. In an ecosystem, the biomass decreases from the producer level to the consumer level. Pyramid of biomass in a grassland ecosystem. The biomass of grasses is the maximum. The biomass gradually decreases towards the consumer level in the following order

3. Pyramid of Energy

The energy flows in an ecosystem from the producer level to the consumer level. At each trophic level, 80 to 90% of energy is lost. Hence, the amount of energy decreases from the producer level to the consumer level. This is represented in a pyramid of energy.

There is always a gradual decrease in the energy content at successive trophic levels from producers to various consumers. This is because on each transformation of energy, there is energy loss in the form of heat.

UNIT -III

ENERGY RESOURCES

ENERGY:

Energy is defined by physicists as the capacity to do work. Energy is found on our planet in a variety of forms, some of which are immediately useful to do work, while others require a process of transformation. The sun is the primary energy source in our lives. Besides, water, fossil fuels such as coal, petroleum products, water, nuclear power plants are sources of energy.

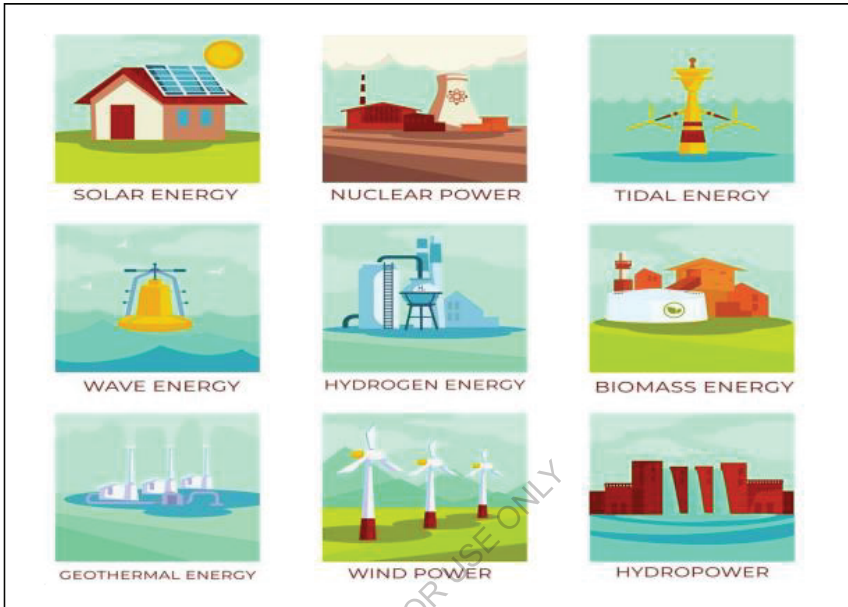
CLASSIFICATION OF ENERGY:

***Renewable Energy Resources**

***Non- Renewable Energy Resources**

Renewable Energy Resources:

Renewable energy systems use resources that are constantly replaced and are usually less polluting. Examples include hydro power, solar, wind, and geothermal (energy from the heat inside the earth). We also get renewable energy from burning trees and even garbage as fuel and processing other plants into bio-fuels.



1.Solar energy:

It is the biggest source of Renewable energy. The solar energy is trapped and converted into thermal and electrical energy. The solar energy is used in devices such as solar cookers, solar heaters and solar batteries. The solar energy is also used for community lighting pumping of water for irrigation and drinking purposes.

2. Hydel power

The energy of running water is used to generate electricity in hydel power stations. The bhakranangal dam is one of the most important hydel power stations of our country. Presently the hydel power generation in the country is about 23,800MW.

3. Wind energy

The energy is used in certain parts of our country to run windmills. These windmills in turn are used to generate electricity adherence, to pump water. Speed of wind is very high in coastal areas therefore, such places are good areas for harnessing wind energy. Uris in Orissa and Tuticorin in Tamil nadu are harnessing wind energy to generate electricity. More than 1000MW electricity is being generated in India by wind.

4. Tidal power

The energy from sea waves or tides is used in some parts of the world to generate hydroelectric power. India has a good potential to build tidal power stations in Gujarat and west Bengal. This source is inexpensive and inexhaustible. However, research and development is necessary for exploitation of tidal energy at affordable cost.

5. Biogas (Go-bar gas)

It is a cheap and convenient cooking fuel. Garbage, crop residue, sewage, cow dung and other waste materials form biomass.

Biomass is decomposed in the absence of oxygen to produce methane gas, which can be used as a fuel, and the residue can be utilized as manure for plants. The gas generated can be used for cooking as well as lighting houses and streets.

NON-RENEWABLE ENERGY RESOURCES

A non-renewable resource (also called a finite resource) is a natural resource that cannot be readily replaced by natural means. An example is carbon-based fossil fuel. The original organic matter, with the aid of heat and pressure, becomes fuel such as oil or gas. Earth minerals and metal ores, fossil fuels (coal, petroleum, natural gas) and groundwater in certain aquifers are all considered non-renewable resources.

1. Petroleum

Petrol, diesel and kerosene are obtained from petroleum. For household purposes burning fuels such as kerosene, oil, and LPG obtain energy. Petrol and diesel are used in cars, scooters, aeroplane, motors and other vehicles as fuels to produce energy.

2. Coal

Coal is the most abundant fossil fuel in the world, and the history of coal goes as far back as 250 millions of years. The plants, which were buried millions of years ago, were converted into hard coal due to the pressure and heat. Coal is used to generate electricity in thermal power stations. Thermal power stations have been established in India at several places.

3. Natural gas

Natural gas is a very effective and useful fuel. It is mainly composed of methane (95%) and small amounts of propane and ethane. The simplicity of natural gas and its low sulphur content makes it the least pollute source of energy.

4. Nuclear power

Nuclear power is the energy stored in the nucleus of an atom. When the nuclei of radioactive elements, like uranium and thorium split, they give out large quantity of heat energy. This converted into electricity. In India, nuclear power stations are located at tarpaper (Maharashtra), Kota (Rajasthan), Kalpakkam (Tamilnadu) and Narora (UP).

CUBER WASTE AND ITS MANAGEMENT:

Any useless, unwanted, or discarded material that is not a liquid or a gas is classified as solid waste. It is old newspaper, bottles, cans, worn out appliances, furniture, abandoned vehicle, flash, Industrial wastes, etc.

SOLID WASTES CAN BE BROADLY CLASSIFIED AS FOLLOWS:

1. Domestic wastes: wastes from household preparation, cooking, cloth, Plastics, etc.
2. Municipal wastes: Garbage and rubbish from houses, institutions, etc.
3. Industrial wastes: packing materials, tannery waste, metal scraps, etc.

4. Agricultural waste: waste from farms wastes after the harvested.
5. Special waste: Hazardous wastes from nuclear power plants, laboratories, hospitals, etc.
6. E-Waste (Electronic and electrical wastes): Electronic equipment like computers, printers, mobile phones, Xerox, machines, calculators, etc., after using these instruments, they are thrown as waste.

DISPOSAL OF SOLID WASTE:

1. Open dumps

People often casually throw solid wastes such as empty bottles, cans, packing materials in the streets, creating visual pollution. In urban areas, the municipal wastes are dumped in certain selected open area and sometimes the dumping site may be a valuable wetland. Open dumping supports disease – carrying insects and rodents. The decomposition causes bad odor also.

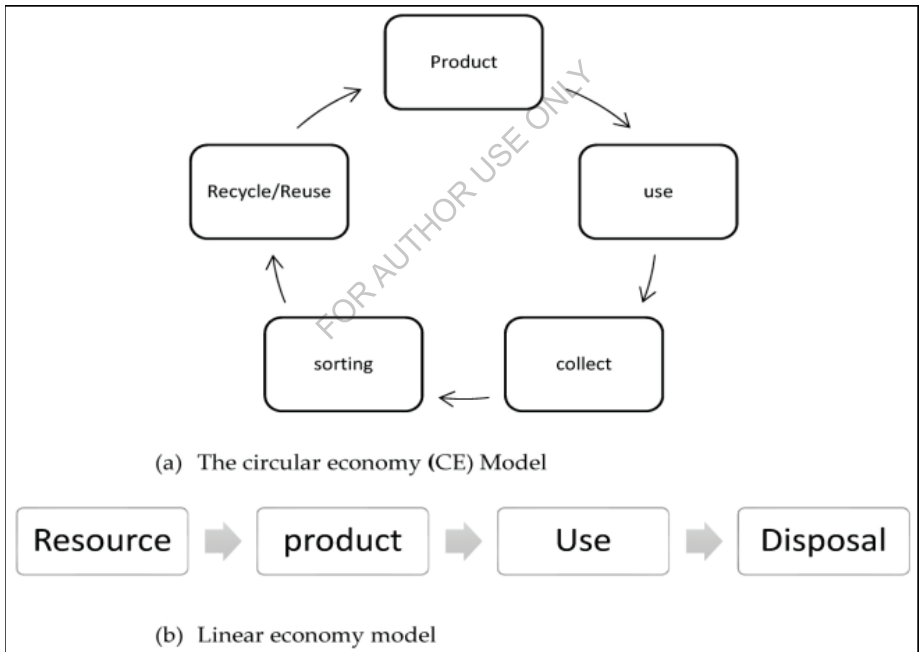
2. Sanitary landfills

A sanitary landfill is a land waste disposal that eliminates most of the problems associated with open dumps by spreading waste in thin layers, compacting them, and covering them with a fresh layer of soil each day.

3. Incineration

Another way to deal solid waste is incineration, the burning of combustible materials and melting of certain non-combustible materials and municipal incinerators. However, even with air pollution devices, incinerators emit large quantities of fine particulate matter into the atmosphere.

WASTES PRODUCTS RECYCLING;



The waste products can be utilized by the following three techniques.

1. Reuse: a given waste material can be used for multiple purposes.
2. Reclamation: A component of the waste product is recovered for use in a manner different from its original
3. The waste products can be utilized by the following three techniques.
4. Reuse: a given waste material can be used for multiple purposes.
5. Reclamation: A component of the waste product is recovered for use in a manner different from its original use.
6. Recycling: Conversion of resources is generally known as recycling. In recycling, a product at the end of its service life is converted into another useful product. Recycling involves separating collecting, processing, marketing and ultimately using material that could have been thrown away. Ash paper can be recycled to another sheet (or) other paper products, cans, bottles and pouches can be recycled for other uses.

Example;

1. Clean water from wastewater treatment can be reused. Refilling of used cool drinks bottles.
2. Skins are used for leather production.
3. Waste paper is used to produce new paper, card boarded.
4. Fly ash is utilized in brick chamber as a substitute for cement.
5. Plastic waste products are recycled to make plastic products, soft waxes, greases and adhesives, etc.

UNIT – IV

NATURAL RESOURCES AND POLLUTION:

Natural resources are resources that exist without actions of humankind. This includes all valued characteristics such as magnetic, gravitational, and electrical properties and forces. On earth it includes: sunlight, atmosphere, water, land along with all vegetation and animal life that naturally subsists upon or within the heretofore identified characteristics and substance. With the world's population fast approaching almost 8 billion, the issue becomes one of resource management as well. It is one thing to practice conservation, but we must understand that as the population grows the demand grows as well. Another problem is that as 3rd world countries continue to progress in industrialization, their need for resources is going to increase. It isn't just that you can plan for an increase in need according to an increase of the percentage of use per billion persons; as societies grow and become more modernized, their reasonable use of natural resources also increases in proportion to people and industry.

TYPES OF NATURAL RESOURCES:

1. Water

While the earth may be mostly water, only about 2-1/2 percent of it is freshwater. Of that 2.5%, even less is considered to be potable.

Potable water is water that is considered to be safe to drink and cook with. Although many countries are working to build water treatment plants, the facts that due to changes in climate, the amount of rain and ice melts from winter have dropped off and lowered the reserve supplies of freshwater to be treated.

There are initiatives to educate and regulate the use of water in the world, as well as exploration into the technology of water farming in arid countries too.

2.Minerals

Other minerals like gypsum, phosphate, bentonite, mica, titanium, and zirconium are found in sea beds along the coastal plains. Rare earth elements like scandium and terbium are two of the earth's powerful minerals that are used in wind turbines and electronic circuits in smart phones. Coastal plains contain deposits of potassium carbonate and rare elements like cerium and neodymium.

3.Forest

As the world gets more modern and the population grows there is more demand for housing and construction projects. This reduces open green spaces. Forests are necessary to preserve the ecology of the world that supports all of the natural resources and life. Forests provide habitat for innumerable creatures on earth, play a critical role in providing clean air and the lumber that builds the homes. Forestry is a renewable industry as long as we manage this resource properly. The forestry industry is huge in countries around the globe, such as Canada, a country with a strong dependence on the forestry industry. Apart from the ones listed above, rocks and sediments, rivers and lakes, mountains, farmland, wetlands, coastal shores, clays, sand, copper, manganese and stone are some of the other natural resources.

4.Livestock

Livestock sector is an important component of India's economy in terms of income, employment, equity and foreign exchange earnings. Among the livestock sector, cattle and buffalo play a major role in India's economy since it has about 13% of world cattle population and 56% (105 million) of world buffalo population. (Table 1). These animals are maintained by rural families with small land holdings of 1 - 2 acres and landless laborer for milk, draught and meat production. In addition, India has 140.5 million goats, 71.5 million sheep and 11 million pigs for meat and fiber production.

5. Animal

For humans, animals are a productive resource. First of all, they supply a wide variety of foods that man needs to survive: milk, cheese, eggs, butter, salami and cold meat, etc. Some animal species, such as corals and oysters, are used by man to produce jewels and handicrafts. The hide of some animals is used instead to produce clothes. In many places across the world, animals are still one of the main means of transport. For instance, camels are particularly used in such desert areas as the Sahara desert, while in Northern areas sleighs are drawn by huskies, dogs that are very resistant to cold and fatigue. Every day, animals join man and help him in the most diverse activities. For their innate sensory superiority and their great adaptability to the external climatic conditions, they can carry out tasks that man cannot.

ENVIRONMENTAL POLLUTION:

Environment pollution may be defined as, “the unfavorable alteration of our surroundings”. It changes the quality of air, water and land, which interferes with the health of humans and other life on earth. Pollution is of different kinds depending on the nature of pollutant generated from different sources.

For example

Automobiles, Industry, nuclear reactors, generate different types of pollutants causing pollution to air, water bodies and land.

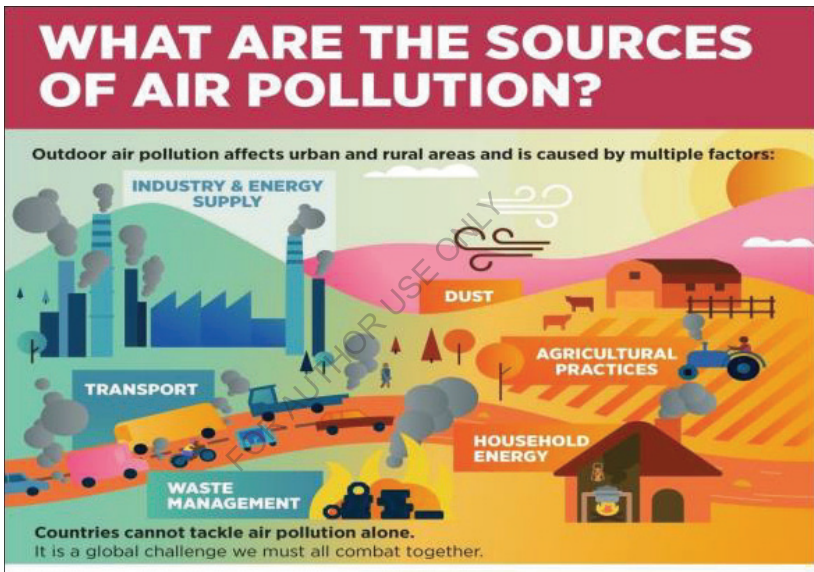
Classification of pollution

The different kinds of pollution that affects the environment are Air pollution, Water pollution, Soil pollution, Marine pollution, Noise pollution, and Thermal pollution and Nuclear hazards.

Air pollution

Definition

Air is a mixture of various gases such as nitrogen, oxygen and carbon dioxide, etc. All these gases are present in a particular ratio. If some other substances are added to air then an imbalance is caused in the ratio of various gases and we say that air is polluted.



SOURCES OF AIR POLLUTION:

All the sources of air pollutants can be classified into two groups: Natural sources, and manmade sources.

1. Natural sources: Dust storms, Forest fires, Volcanoes, Pollen grains

2. Man-Made sources

- (i) Burning of fuels like wood, coal and kerosene in homes pollute the air.
- (ii) Exhaust gases emitted by motor vehicles pollute the air.
- (iii) Industries pollute air by releasing various types of pollutants.
- (iv) Nuclear power plants pollute air by releasing radioactive rays.
- (v) Deforestation pollutes air by increasing the amount of carbon dioxide
- (vi) Use of Chlorofluoro carbons (compounds used in refrigeration, fire extinguishers and aerosol sprayers) pollutes air by depleting ozone layer.
- (vii) Mining of coal releases coal dust into air and pollute it.

Harmful Effects of Air pollution

1. The poisonous gases as sulphur dioxide and nitrogen oxide present in polluted air causes difficulty breathing and result in diseases such as lung cancer and asthma.
2. Air pollution causes, green house effect that forms excessive heating of the earth and its atmosphere.
3. Air pollution depletes the ozone layer due to which ultra-violet radiations can reach the earth and cause damage.
4. Air pollution can cause acid rain which damage crop plants, trees, buildings, statues and metal structures.

PREVENTION AND CONTROL OF AIR POLLUTION:

1. By installing tall chimneys in homes and factories.
2. By growing more trees along road-sides and houses.

3. By using smokeless fuels like LPG and natural gas in place of coal, wood and Kerosene oil for cooking purposes. By installing electrostatic precipitations in chimneys of industries.
4. By cleaning the polluted air by passing through fabric filters or wet scrubbers.

Water pollution

Definition

The contamination of water of rivers and lakes with unwanted and harmful substances is called water pollution.



SOURCES OF WATER POLLUTION:

1. **Industrial wastes:** The discharge of industrial wastes from industries into river and takes polluter water. This waste contains harmful chemicals including acids and alkali.
2. **Sewage:** Dumping of sewage into rivers and lakes polluter water.
3. **Chemical fertilizers:** The excessive use of fertilizers in the soil causes water pollution.
4. **Synthetic detergents:** Modern day detergents used for brighter and quicker washing degrade slowly. They accumulate in water rendering it unfit for human consumption.
5. **Petroleum oil:** The leakage of petroleum oil into the sea during drilling and shipping operations polluter the water.

HARMFUL EFFECTS OF WATER POLLUTION:

1. The polluted water contains a number of disease causing pathogens like bacteria and virus, which cause water-borne diseases like typhoid, cholera, dysentery and jaundice.
2. The acid and alkali present in industrial wastes kill those microorganisms, which are the natural cleaning agents of water.
3. The sewage and fertilizers present in polluted water are rich in nutrients. This leads to an excessive growth of algae, which cover the whole surface of lake. When the algae die, the Decomposers use up all the dissolved oxygen for decomposing the vast amount of dead algae. Since no dissolved oxygen is left in the water of the lake, all the aquatic animals like fish die.
4. The industrial wastes of industries like paper, paint, pesticides, metal etc. Contain toxic chemicals like mercury, lead, copper, cadmium, etc., which can enter the human body through food chains and affect their health.

PREVENTION AND CONTROL OF WATER POLLUTION:

1. Sewage should be treated suitably before it is discharged into the river.
2. Strict legislation should be enacted to make it obligatory for the industries, to treat their wastewater before it is discharged into river, or seas.
3. Solid factory waste, containing organic matter should be sent to biogas plants for generation of energy.
4. Waste food material, paper, decaying vegetables and plastics should not be thrown into open drains.

Noise pollution

Definition

A loud unpleasant sound or an unwanted sound of any kind is called noise pollution. It adversely affects our physical and mental health.



Sources of noise pollution:

1. Various industries such as textile , mills, printing press, engineering establishment, etc.
2. Agricultural machines like tractors, harvesters, etc.
3. Defense equipment such as tanks, rocket launching, shooting practices,etc.
4. Entertaining equipment like radio, T.V. record PlayerNet.
5. Domestic gadgets such as desert coolers, fans, mixers, pressure cooker,etc.
6. Public address systems like loudspeaker.
7. Transport vehicles like scooters, cars, buses, jet planes.

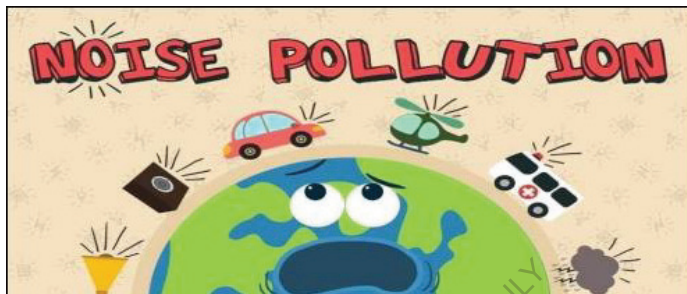
Effects of Noise pollution:

1. Prolonged exposure to high intensity of noise can lead to permanent loss of hearing ability.
2. Prolonged noise pollution causes damage to heart, brain and liver in animals.Noise causes emotional disturbances, headache and lack of concentration.
3. Noise causes increased heartbeat and blood pressure. A sudden loud noise can lead to heart failure.

Control of Noise pollution:

1. Noise producing industries, railway stations, airports should be located away from residential areas. The industrial machines and engines of motor vehicles should be maintained properly and lubricated from time to time.

2. Traffic police personnel and factory workers exposed to high noise pollution should use earplugs or muffs. The noise of motor vehicles on the road can be reduced by planting many rows of trees.
3. There should be silence zones near residential areas, educational institutions and hospitals.



Soil (or Land) pollution

Definition

Soil pollution may be defined as “The contamination of the soil resulting in either reduction of its fertility or entry of contaminants in plants which subsequently enter the food chain and impart toxic or detrimental effect on the consumers”.



Sources of soil pollution:

The main sources of soil pollution are solid wastes, acid rain, excessive use of fertilizers and use of insecticides and herbicides.

1. **Solid wastes:** A lot of soil pollution is caused by solid wastes. This solid waste comes from homes, agricultural fields, industries and many other places.
2. **Acid Rain:** Acid rain is caused by air pollution. This acid rain falls to the soil and pollutes it.
3. **Fertilizers:** The excessive use of fertilizers makes the soil either acidic or alkaline and pollutes it.
4. **Insecticides and Herbicides:** These chemicals are used to check the growth of harmful insects and herbs. Excessive use of insecticides or herbicides pollutes the soil.

Harmful Effects of soil pollution:

1. Soil pollution caused by acid rain reduces the fertility of soil, which in turn affects the crop yield.

2. Soil pollution caused by the excessive use of fertilizers reduces the fertility of the soil making it acidic or alkaline.
3. The soil pollution caused by insecticides or herbicide is very harmful because these chemicals can enter our body through food chains and damage our health.

Prevention of soil pollution:

1. Reuse

Reuse and recycle of non-biodegradable wastes is the best way to reduce soil pollution. Recycling is the reprocessing of discarded materials into new useful products.

2. Composting

The biological wastes can be converted into a useful product through composting biological degradation or breakdown of organic matter under aerobic conditions.

3. Energy from wastes

Organic wastes can be decomposed in large. Oxygen free digesters to produce methane under controlled conditions. In us, some farmers produce all the fuel they need by generating methane from animal manure.

4. Landfills

Solid wastes especially non-biodegradable are regulated and controlled in landfills to decrease smells and litter and to discourage insect and rodent population. Landfills are being built away from rivers, lakes, flood plains and aquifer recharge zones.

Environmental law and acts in India

‘Environment’ is a very comprehensive term. It includes within its ambit a wide variety of phenomenon. It is a dynamic term that may be used to describe a limited area on one hand, and the entire planet on the other. The term Environment may be perceived in different connotations. There are numerous definitions of the term as provided by different National and International legal instruments. Generally speaking, Environment includes the external conditions, resources, stimuli etc. with which an organism interacts. The Preamble of the United Nations Declaration on Human Environment, adopted in Stockholm in June 1972 states, “Man is both creature and molder of his environment, which gives him physical substance and affords him the opportunity for intellectual, moral, social and spiritual growth”. The environment is clearly at risk from a variety of sources of harm, mostly of human origin. In order to tackle this problem it is important that we develop strategies for modifying human behavior towards environmentally benign practices and away from environmentally damaging ones. In very broad terms, techniques for modifying human behavior can be thought of as falling into two types: incentives and disincentives. Law is important as it creates a framework within which incentives and disincentives can operate. Law is all pervasive. Other methods for influencing human behavior are to a certain extent, voluntary or optional. Education, ethics, peer and family pressure: these all apply in various degrees. Law, on the other hand, cannot easily be avoided. It is axiomatic to the “rule of law” that law in a society applies equally to everyone at all-time.

Environmental procedures in India

Recognizing the importance of Environmental Impact Assessments in sitting development activities in an environmentally sustainable manner, the Ministry of Environment and Forest, Government of India has prescribed a procedure for restricting the expansion and modernization of activities being undertaken in any part in India unless environmental clearance has been accorded by the Moe f, Government of India or in certain cases by the State Government. This was earlier through the E.I.A. notification of 1994 and its various amendments. The E.I.A. notification of 1994 as above was superseded and replaced by notification no S.O. 1533 dated 14-09-2006 issued by the Government of India. This notification provided for the establishment of the State Environmental

Impact Assessment Authorities, the Central and State Expert Appraisal Committees and the procedures for granting environmental clearances. It is under this notification that clearances are issued by the MoEF after 2006. The Environment Impact Assessment notification of 2006 identifies 37 activities in 8 categories of projects and activities (New projects and expansion or modernization of existing projects) where prior Environmental clearance must be obtained by the project proponents (Schedule 1). The activities are classified into two categories A and B. Category A projects are considered through an Expert Appraisal Committee constituted at the level of the Central Government and granted/refused a clearance by the MoEF and CC, GOI on the recommendation of this committee. Category B projects are considered through the State Expert Appraisal Committees and granted /refused clearances by the State Environmental Impact Assessment Authority. In the absence of a duly constituted authority/committee at the State Level, a category 'B' project automatically becomes a category 'A' project. Screening, Scoping, public consultation (with exceptions) and appraisal constitute the various stages of project evaluation and time frames for each stage prescribed. Environmental Impact Assessments have been provided as an essential management tool (with exceptions) for the clearance mechanism. Quality Council of India, accredited consultants is authorized to make presentations on the EIA and EMP before the respective Appraisal Committees.

A six monthly monitoring schedule is also prescribed. Monitoring is done by the Ministry of Environment, Forest and Climate Change. This is essentially a clearance granted by the Ministry of Environment and Forests and the SEIAA/SEAC represent constituent entity of the Ministry at the State Level. Validity of Environmental Clearances is generally limited to 10 years and the Environmental Clearance is transferable. This clearance is different from the consents to establish issued by the State Boards and Committees and the conditions of the Environmental Clearance are generally included as condition of consent aforementioned.

Social Issues:

Environmental issues are harmful biophysical environment. Environmental protection is a practice of protecting the natural environment on the individual, organizational or governmental levels, for the benefit of both the environment and humans. Environmentalism, a social and environmental movement, addresses environmental issues through advocacy, education and activism.

The carbon dioxide equivalent of greenhouse gases (GHG) in the atmosphere has already exceeded 400 parts per million (NOAA) (with total "long-term" GHG exceeding 455 parts per million) (Intergovernmental Panel on Climate Report). The amount of greenhouse gas in the atmosphere is possibly above the threshold that can potentially cause climate change. The UN Office for the Coordination of Humanitarian Affairs (OCHA) has stated "Climate change is not just a distant future threat. It is the main driver behind rising humanitarian needs and we are seeing its impact. The number of people affected and the damages inflicted by extreme weather has been unprecedented."Further, OCHA has stated:

- Climate disasters are on the rise. Around 70 percent of disasters are now climate-related – up from around 50 percent from two decades ago.
- These disasters take a heavier human toll and come with a higher price tag. In the last decade, 2.4billion people were affected by climate-related disasters, Compared to 1.7 billion in the previous decade. The cost of responding to disasters has risen tenfold between 1992 and 2008.
- Destructive sudden heavy rains, intense tropical storms, repeated flooding, and droughts are likely to increase, as will the vulnerability of local communities in the absence of strong concerted action.
- Environment destruction caused by humans is a global, ongoing problem. By the year 2050, the global human population is expected to grow by 2 billion people, thereby reaching a level of 9.6 billion people.
- The human effects on Earth can be seen in many different ways. A main effect is an increase in global temperature. According to the report "Our Changing Climate", the global warming that has been going on for the past 50 years is primarily due to human activities.
- Since 1895, the average temperature in the United States has increased by between 1.3°F and 1.9°F, with most of the increase taking place since around 1970.

Development

In India, pollution and environmental degradation have reached alarming dimensions due to poverty, deforestation, industrial development without adequate environmental safeguards, and sheer greed. Fortunately, public concern, rooted in the country's past, has revived. Major pollutants and critically affected areas have been identified. Pollution control of water, air, and land has been established by both official and private organizations and the work on environmental protection is steadily growing. The Ganga purification plan is a representative case study. Poverty alleviation is a long term process. It is India's major problem and is being tackled with help from private enterprise and by international assistance. Simultaneously, environmental protection through pollution control, is also receiving administrative and legislative support and fiscal assistance through direct and indirect tax incentives. The country's courts are rendering valuable help to environmentalists by pronouncing far-reaching decisions in public-interest litigation. To boost the existing environment-protection movement, greater emphasis is urgently needed for environmental education, peoples' participation, population control, and cost-effective pollution control measures.



UNIT – V

BIODIVERSITY AND INDIA:

Introduction

Bio means 'life' and diversity means 'variety', hence, biodiversity refers wide variety of life on the earth.

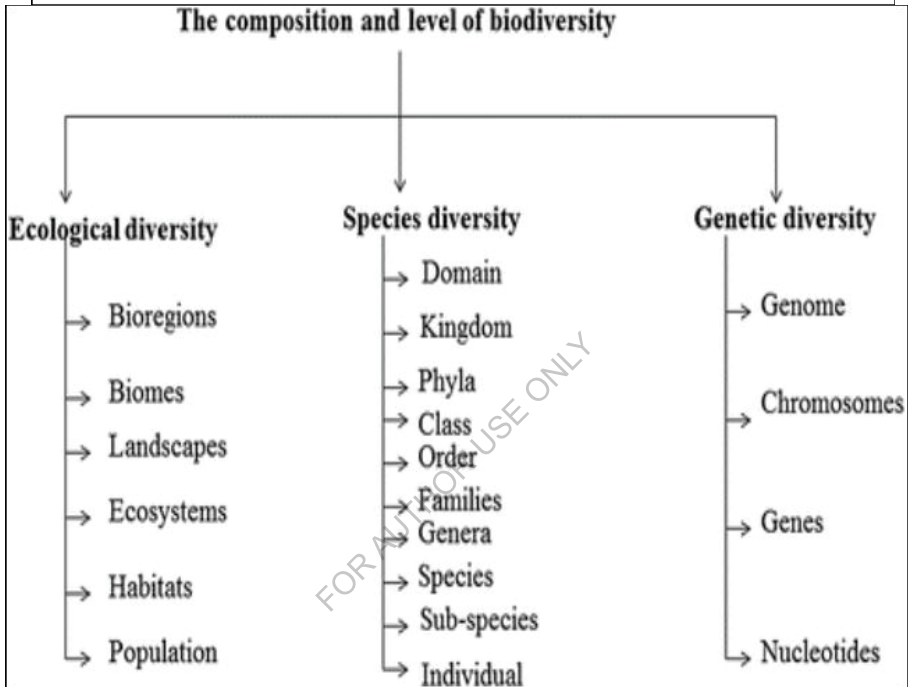
Our planet-earth (biosphere) contains more than 20 million species of organisms. However, of which only 1.4 million species have been identified so far. These species differ widely from one another. This variation in living organisms is called biodiversity.

The biodiversity is an index of vegetation process. Bio diversities include the variety of living organism, the genetic differences among them and the communities and ecosystem in which they occur.

Significance of Biodiversity

- Biodiversity is very important for human life, as we depend on plants, microorganisms, earth's animals for our food, medicine and industrial products.
- Biodiversity protects the fresh air, clean water and productive land.
- It is also important for forestry, fisheries and agriculture, which depend on rich variety of various biological resources available in nature.
- Loss of biodiversity has serious economic and social costs for any country.

Types of biodiversity



Biodiversity is generally classified into three types are genetic diversity, species diversity and community (or) ecosystem diversity.

1. Genetic diversity

Genetic diversity is the diversity within species. Genetic diversity is the combination of different genes found within a population of a single species, and the pattern of variation found within different populations of the same species.

Example

There are number of teak wood varieties found available. That is Indian teak, Burma teak, Malaysian teak, etc.

2. Species diversity

A discrete group of organisms of the same kind is known as species. Species diversity is the diversity between different species. The sum of varieties of all the living organisms at the species level is known as species diversity.

Example

Total numbers of living species in the earth are about more than 20 million. However, of which only about 1.5 million living organisms are found and given scientific names.

Plant species: Apple, mango, grapes, wheat, rice, etc. ***Animal species:*** Lion, tiger, elephant, deer, etc.

3. Community (or) Ecosystem diversity

It is a set of biotic components (plants, animals and microorganisms) interacting with one another and with biotic components (soil, air water, etc) .The diversity at the ecological or habitat level is known as ecosystem diversity. A large region with different ecosystem can be considered as ecosystem diversity.

Example: River ecosystem

The river, which include the fish, aquatic insects, mussels and variety of plants that have adapted.

Bio diversity at Global Level

Biodiversity is the measure of the variety of earth's animal, plant and microbial species; of genetic differences within species; and of the ecosystems that support the species. Out of an estimated 30 million species on earth, only one-sixth has been identified and authenticated in the past 200 years.

An estimated biodiversity covers 400,000 higher plants. Most of the world's bio-rich nations are in the South, which are the developing nations. In contrast, the majority of the countries capable of exploiting biodiversity are Northern nations, in the economically developed world.

Bio diversity at National Level

India is 10th among the plant rich countries of the world, fourth among the Asian countries, eleventh according to the number of endemic species of higher vertebrates (amphibian, birds and mammals), and tenth in the world as far as richness in mammals is concerned. Out of the 10 'Hot spots' identified in the world, India has four. These are Eastern Himalaya, North East India, Western Ghats and Andaman & Nicobar Islands. The crops which first grew in India and spread throughout the world include rice, sugarcane, Asiatic viginas, jute, mango, citrus, and banana, several species of millet, spices, medicinal, aromatics and ornamental. India ranks sixth among the centers of diversity and origin in terms of Agro-biodiversity.

Loss of biodiversity

1. The farmers prefer hybrid seeds, as a result, many plant species become extinct.
2. For the production of drugs, the pharmaceutical companies collect wild plants, so several medicinal plants now become extinct.
3. Tropical forest is the main sources of world's medicine. Every year these forests are disappearing due to agriculture, mining and logging.

Cause of Biodiversity

The main cause of the loss of biodiversity can be attributed to the influence of human beings on the world's ecosystem, In fact human beings have deeply altered the environment, and have modified the territory, exploiting the species directly, for example by fishing and hunting, changing the bio Geo chemical cycles and transferring species from one area to another of the Planet.

The threats to biodiversity can be summarized in the following main points:

- ***Alteration and loss of the habitats:*** the transformation of the natural areas determines not only the loss of the vegetable species, but also a decrease in the animal species associated to them.
- ***Introduction of exotic species and genetically modified organisms:*** species originating from a particular area, introduced into new natural environments can lead to different forms of imbalance in the ecological equilibrium. Refer to, “Introduction of exotic species and genetically modified organisms”.
- ***Pollution:*** human activity influences the natural environment producing negative, direct or indirect, effects that alter the flow of energy, the chemical and physical constitution of the environment and abundance of the species;
- ***Climate change:*** for example, heating of the Earth’s surface affects biodiversity because it endangers all the species that adapted to the cold due to the latitude (the Polar species) or the altitude (mountain species).
- ***Over exploitation of resources:*** when the activities connected with capturing and harvesting (hunting, fishing, farming) a renewable natural resource in a particular area is excessively intense, the resource itself may become exhausted, as for example, is the case of sardines, herrings, cod, tuna and many other species that man captures without leaving enough time for the organisms to reproduce.

Remedial Measures

- Dealing with biodiversity loss is tied directly to the conservation challenges posed by the underlying drivers.
- Conservation biologists note that these problems could be solved using a mix of public policy and economic solutions assisted by continued monitoring and education.
- Governments, nongovernmental organizations, and the scientific community must work together to create incentives to conserve natural habitats and protect the species within them from unnecessary harvesting, while distinctiveness behavior that contributes to habitat loss and degradation.

- Sustainable development (economic planning that seeks to foster growth while preserving environmental quality) must be considered when creating new farmland and human living spaces.
- Laws that prevent poaching and the indiscriminate trade in wildlife must be improved and enforced. Shipping materials at ports must be inspected for stowaway organisms.

Hotspot of biodiversity

A biodiversity hot spot is a bio geographic region that is both a significant reservoir of biodiversity and is threatened with destruction. The term biodiversity hot spot specifically refers to 25 biologically rich areas around the world that have lost at least 70 percent of their original habitat.

Cool spot of biodiversity

There are cool-spots (the world's last refuges where high numbers of threatened species still persist). Cool-spots could be the result of protection or because of intact habitat that has not been cleared yet. India still has crucial refuges that need protecting.

Conservation of biodiversity

Conservation of biodiversity can be achieved in a number of complementary ways. These methods fall within the broad concept of 'Gene banks'. They can be classified into the following types.

(1) In situ conservation

This includes conservation of plants and animals in their native ecosystems (or) even in synthetic ecosystems. This type of conservation applied only to wild fauna and flora and not to the domesticated animals and plants, because conservation is achieved by protection of population in nature. It includes system of protected areas. National Park, Sanctuaries, Biosphere reserved.

(2) Ex-situ conservation

Which includes conservation of samples of genetic diversity-particularly representing, endangered species-away from their habitats.

Ex-situ conservation is done through establishment of gene banks which include genetic resources centers zoos, botanical gardens, culture collections etc. it also includes in-situ conservation methods which include national parks, and sanctuaries which is important for the conservation of crop varieties and world genetic resources.



Save OUR Earth!



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